

SUPPLEMENTARY MATERIAL

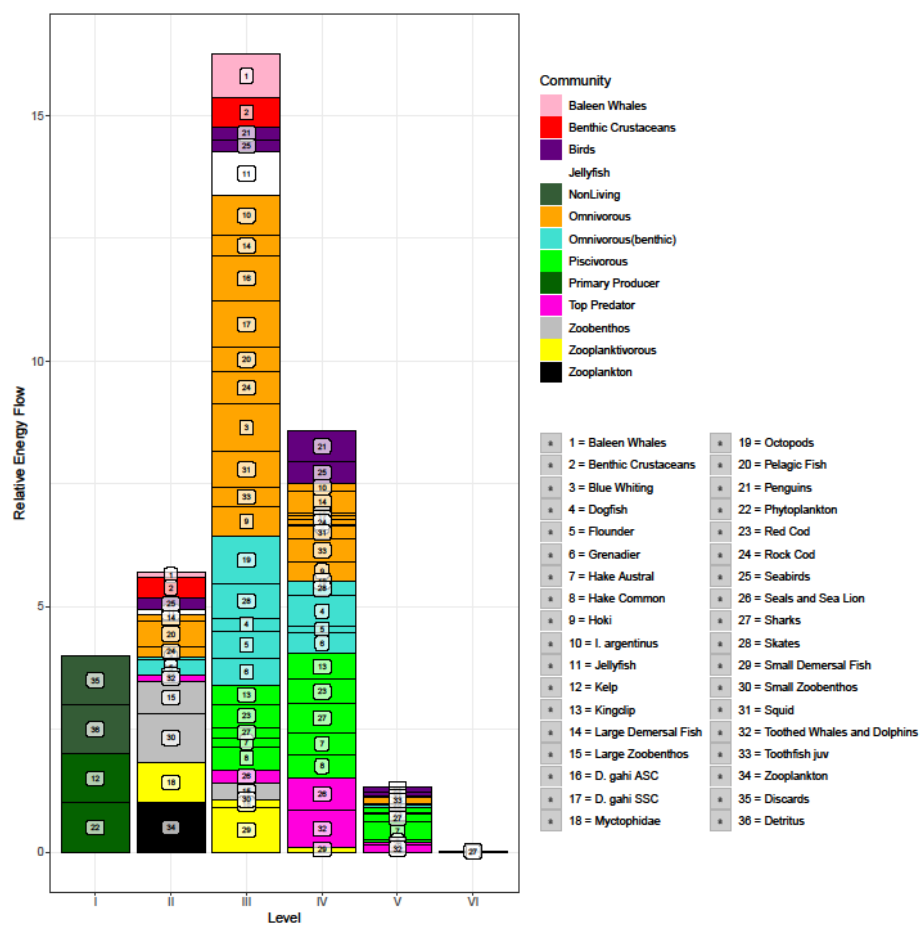


Figure S1: Trophic level decomposition of functional groups of the Ecopath model with relative flow of each group per trophic level (available as interactive plot)

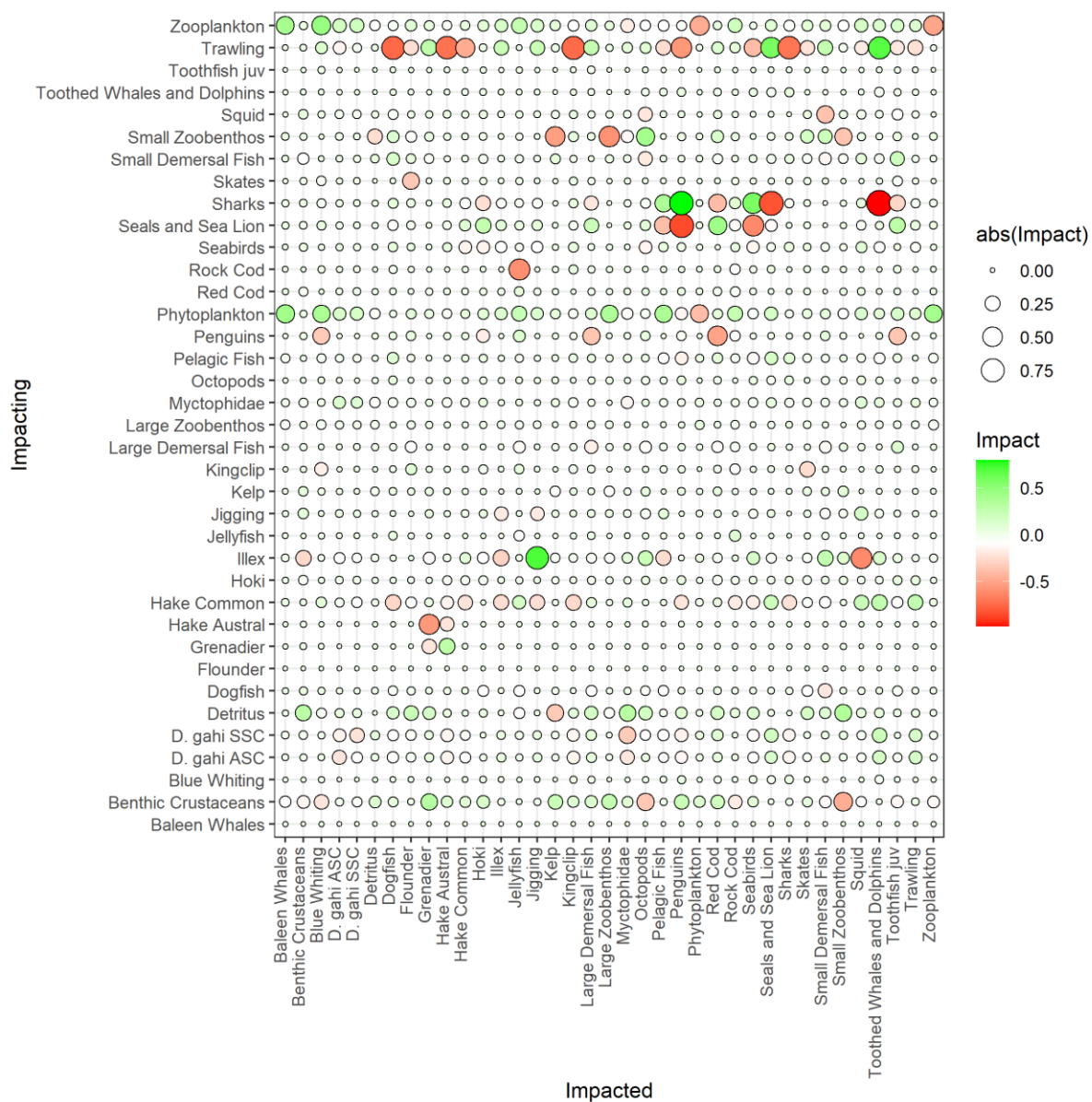


Figure S2: Positive and negative impacts of all groups on each other, derived from the mixed trophic impacts routine from the Ecopath model; green = positive impacts, red = negative impacts, size = absolute impact

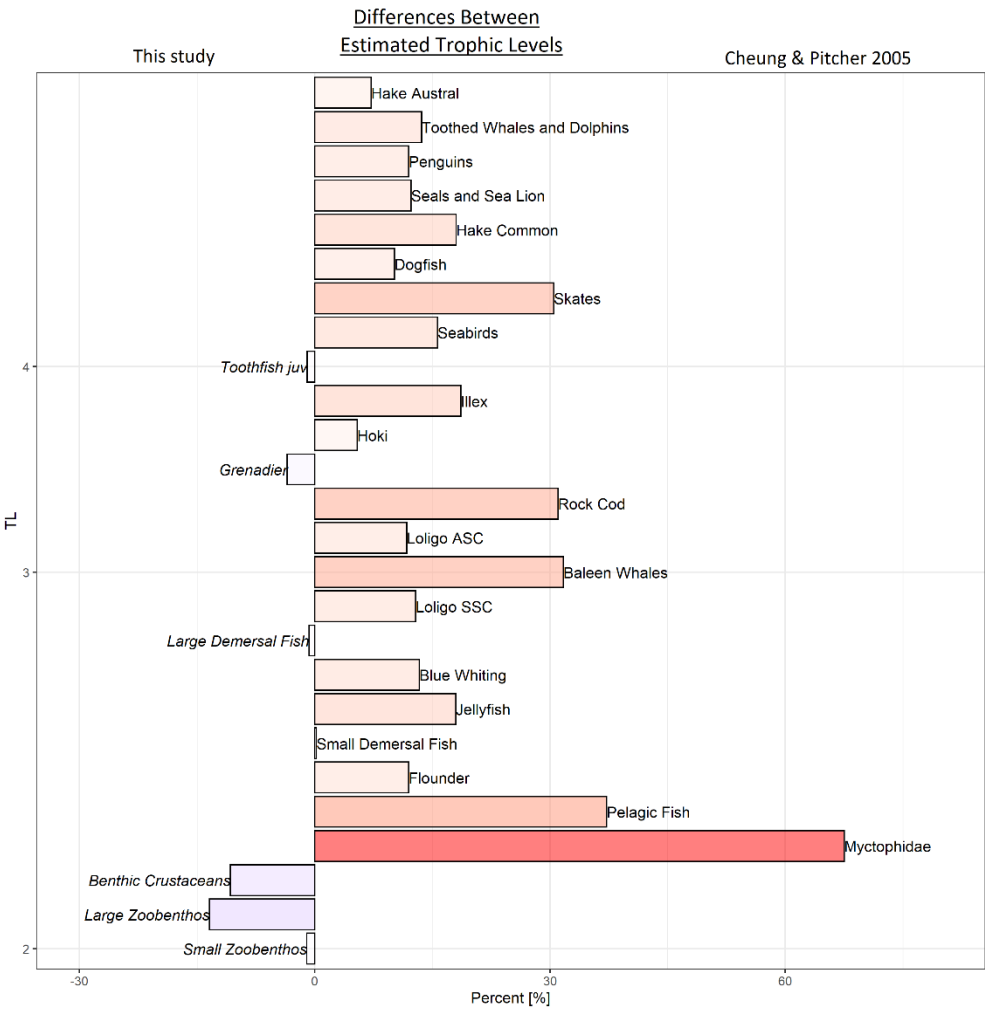


Figure S3: Difference (in%) in trophic level between the Ecopath model (this study) and the model by Cheung & Pitcher (2005). Differences are ordered according to trophic level values (bottom low, top high).

Table S1: Number of samples per month for the stable isotope groups

Mo nth	Benthic M. gregaria	Bir ds	Omniv orous	Omnivorous(benthic)	PO M	Piscivo rous	Primary Producer	Top Predato rs	Zooplankti vorous	Zoopla nkton
1	-	-	108	-	-	-	-	-	-	-
2	5	2	284	2	1	3	3	3	3	13
3	-	5	36	3	-	5	-	-	-	-
4	1	2	103	8	-	7	-	-	-	-
5	-	-	16	5	-	14	-	-	-	-
6	-	1	16	-	-	-	-	-	-	-
7	11	-	316	29	-	34	7	-	6	-
8	-	2	-	-	-	-	-	-	-	-
9	-	12	27	10	-	6	-	-	-	4
10	2	6	112	13	-	5	2	-	-	-
11	-	-	22	6	-	5	-	-	-	2

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17 **Table S2: Diet Matrix (separate file)**

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19 **Table S3: Initial production (P/B), consumption (Q/B) and ecotrophic efficiency (EE) values for the Ecopath model**

Functional Group	P/B	Q/B	EE
Algae	7.526		0.251
Baleen Whales	0.840	10.365	0.308
Bathydemersal	0.045	0.658	
Benthic crustaceans	2.241	10.944	0.937
Blue Whiting	0.399	13.081	0.759
Corals	6.168		0.895
Demersal fish	1.170	4.950	0.303
Detritus			0.219
Discards			0.487
Dogfish	0.412	5.818	0.965
Flatfish	0.952	7.868	0.753
Flounder	0.962	36.222	
Grenadier	1.055	2.344	0.990
Hake austral	0.297	1.385	0.983
Hake common	0.574	2.686	0.859
Hoki	1.297	2.505	0.888
Illex	1.161	4.996	0.747
Jellyfish	7.360	25.851	0.520
Kingclip	0.435	11.923	0.714
Krill	4.219	21.678	0.735
Large bathydemersal fish	0.525	3.419	0.682
large bathypelagic fish	0.300	2.000	0.610
Large Demersal	0.307	29.149	
Large demersal fish	0.799	5.233	0.737
Large pelagic fish	0.500	2.500	0.950
Large zoobenthos	1.668	10.922	0.864
Loligo	1.172	4.964	0.904
Myctophidae	1.227	9.420	0.642
Pelagic fish	0.890	7.713	0.500
Penguins	4.000	80.000	0.001
Phytoplankton	124.041		0.429
Red Cod	0.543	2.184	0.854
Rock Cod	0.570	21.321	0.980
Seabirds	1.707	81.816	0.217
Seals and sea lion	0.317	16.954	0.284
Sharks	0.435	3.866	0.590
Skates	0.369	2.740	0.592
small bathydemersal fish	1.320	12.000	0.950
Small demersal	0.770	17.780	0.843
Small pelagic fish	1.218	19.766	0.727
Small zoobenthos	3.303	11.719	0.728

Squid	3.084	16.368	0.817
Toothed whales & dolphins	0.525	10.463	0.136
Toothfish	0.562	3.489	0.950
Zooplankton	25.015	93.617	0.775

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Table S4: Landings and Discards used for the Ecopath model (in t per km²)

	Group name	Landings			Discards		
		Trawling	Jigging	Total	Trawling	Jigging	Total
1	<i>Baleen Whales</i>	-	-	-	0	0	0
2	<i>Benthic Crustaceans</i>	-	-	-	0.0002	-	0.0002
3	<i>Blue Whiting</i>	0.0003	-	0.0003	-	-	-
4	<i>Dogfish</i>	-	-	-	0.0004	-	0.0004
5	<i>Flounder</i>	-	-	-	-	-	-
6	<i>Grenadier</i>	0.0006	-	0.0006	0.0003	-	0.0003
7	<i>Hake Austral</i>	0.0003	-	0.0003	-	-	-
8	<i>Hake Common</i>	0.2172	-	0.2172	0.0004	-	0.0004
9	<i>Hoki</i>	0.0382	-	0.0382	0.0004	-	0.0004
10	<i>Illex</i>	-	0.3451	0.3451	-	-	-
11	<i>Jellyfish</i>	-	-	-	-	-	-
12	<i>Kelp</i>	-	-	-	-	-	-
13	<i>Kingclip</i>	0.0082	-	0.0082	0.0001	-	0.0001
14	<i>Large Demersal Fish</i>	-	-	-	0.0003	-	0.0003
15	<i>Large Zoobenthos</i>	-	-	-	-	-	-
16	<i>D. gahi</i> ASC	0.15	-	0.15	0.0001	-	0.0001
17	<i>D. gahi</i> SSC	0.1547	-	0.1547	0.0001	-	0.0001
18	<i>Myctophidae</i>	-	-	-	-	-	-
19	<i>Octopods</i>	-	-	-	-	-	-
20	<i>Pelagic Fish</i>	0.0038	-	0.0038	0.0001	-	0.0001
21	<i>Penguins</i>	-	-	-	-	-	-
22	<i>Phytoplankton</i>	-	-	-	-	-	-
23	<i>Red Cod</i>	0.0071	-	0.0071	0.0001	-	0.0001
24	<i>Rock Cod</i>	0.0013	-	0.0013	0.0027	-	0.0027
25	<i>Seabirds</i>	-	-	-	-	-	-
26	<i>Seals and Sea Lion</i>	-	-	-	-	-	-
27	<i>Sharks</i>	-	-	-	-	-	-
28	<i>Skates</i>	0.0067	-	0.0067	0.0004	-	0.0004
29	<i>Small Demersal Fish</i>	-	-	-	-	-	-
30	<i>Small Zoobenthos</i>	0.0001	-	0.0001	-	-	-
31	<i>Squid</i>	-	-	-	-	-	-
32	<i>Toothed Whales and Dolphins</i>	-	-	-	-	-	-
33	<i>Toothfish juv</i>	0.0014	-	0.0014	-	-	-
34	<i>Zooplankton</i>	-	-	-	0.0001	-	0.0001
35	<i>Detritus</i>	-	-	-	-	-	-
36	<i>Sum</i>	0.5899	0.3451	0.935	0.0059	-	0.0059

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23 **Table S5:** Niche overlap [%] of each functional group and contrasting group on a 95% alpha level, based on $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$
 24 values

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Group	Contrast	Niche overlap prob.
Birds	Mgregaria	13.52
Birds	Omnivorous	65.79
Birds	Omnivorous(benthic)	8.91
Birds	Piscivorous	37.16
Birds	PrimaryProducer	0.04
Birds	Zooplanktivorous	8.28
Birds	Zooplankton	0.54
Mgregaria	Birds	30.27
Mgregaria	Omnivorous	36.87
Mgregaria	Omnivorous(benthic)	51.82
Mgregaria	Piscivorous	22.43
Mgregaria	PrimaryProducer	17.01
Mgregaria	Zooplanktivorous	2.89
Mgregaria	Zooplankton	2.43
Omnivorous	Mgregaria	56.58
Omnivorous	Birds	99.19
Omnivorous	Omnivorous(benthic)	39.04
Omnivorous	Piscivorous	52.72
Omnivorous	PrimaryProducer	0.99
Omnivorous	Zooplanktivorous	60.83
Omnivorous	Zooplankton	8.8
Omnivorous(benthic)	Mgregaria	34.59
Omnivorous(benthic)	Birds	10.76
Omnivorous(benthic)	Omnivorous	14.44
Omnivorous(benthic)	Piscivorous	26.24
Omnivorous(benthic)	PrimaryProducer	0.12
Omnivorous(benthic)	Zooplanktivorous	0.04
Omnivorous(benthic)	Zooplankton	0.01
Piscivorous	Mgregaria	21.52
Piscivorous	Birds	85.14
Piscivorous	Omnivorous	59.02
Piscivorous	Omnivorous(benthic)	38.58
Piscivorous	PrimaryProducer	0.02
Piscivorous	Zooplanktivorous	1.51
Piscivorous	Zooplankton	0.14
Primary Producer	Mgregaria	30.25
Primary Producer	Birds	0.21
Primary Producer	Omnivorous	2.72
Primary Producer	Omnivorous(benthic)	0.65
Primary Producer	Piscivorous	0.12

Primary Producer	Zooplanktivorous	21.68
Primary Producer	Zooplankton	65.16
Zooplanktivorous	Mgregaria	3.27
Zooplanktivorous	Birds	9.54
Zooplanktivorous	Omnivorous	21.15
Zooplanktivorous	Omnivorous(benthic)	0.05
Zooplanktivorous	Piscivorous	1.16
Zooplanktivorous	PrimaryProducer	4.78
Zooplanktivorous	Zooplankton	41.49
Zooplankton	Mgregaria	2.97
Zooplankton	Birds	1.2
Zooplankton	Omnivorous	7.02
Zooplankton	Omnivorous(benthic)	0
Zooplankton	Piscivorous	0.23
Zooplankton	PrimaryProducer	15.25
Zooplankton	Zooplanktivorous	64.09

Table S6: ANOVA Group comparisons $\delta^{15}\text{N}$ versus sex for each species, Df = Degrees of freedom, mean sum of squares, F statistic and p-value

Species		Df	Sum Sq	Mean Sq	F value	p-val
<i>Bathyraja albomaculata</i>	Sex	1	0.151	0.1508	0.166	0.691
	Residuals	12	10.915	0.9096		
<i>Bathyraja brachyurops</i>	Sex	1	0.418	0.4183	0.396	0.535
	Residuals	24	25.375	1.0573		
<i>Cottoperca gobio</i>	Sex	1	0.027	0.0268	0.026	0.875
	Residuals	16	16.713	1.0446		
<i>Dipturus lamillai</i>	Sex	1	0.806	0.8063	1.698	0.215
	Residuals	13	6.172	0.4748		
<i>Dissostichus eleginoides</i>	Sex	1	0.06	0.0597	0.039	0.848
	Residuals	8	12.24	1.5298		
<i>Doryteuthis gahi</i>	Sex	1	0.76	0.7624	0.729	0.394
	Residuals	264	276.23	1.0463		
<i>Genypterus blacodes</i>	Sex	1	0.0111	0.0111	0.033	0.861
	Residuals	7	2.3388	0.3341		
<i>Illex argentinus</i>	Sex	1	5.78	5.775	1.555	0.238
	Residuals	11	40.85	3.714		
<i>Macruronus magellanicus</i>	Sex	1	0.592	0.5923	1.056	0.323
	Residuals	13	7.294	0.561		
<i>Merluccius hubbsi</i>	Sex	1	0.778	0.7785	0.536	0.479
	Residuals	11	15.972	1.452		
<i>Moroteuthopsis ingens</i>	Sex	1	2.532	2.531	1.28	0.278
	Residuals	13	25.711	1.978		
<i>Patagonotothen ramsayi</i>	Sex	1	0.032	0.0318	0.035	0.854

	Residuals	29	26.613	0.9177		
<i>Puffinus griseus</i>	Sex	1	0.0575	0.05752	0.264	0.622
	Residuals	8	1.7458	0.21822		
<i>Salilota australis</i>	Sex	1	1.84	1.842	0.855	0.367
	Residuals	18	38.77	2.154		
<i>Schroederichthys bivirus</i>	Sex	1	6.907	6.907	9.766	0.00653
	Residuals	16	11.316	0.707		
<i>Squalus acanthias</i>	Sex	1	1.79	1.793	0.453	0.52
	Residuals	8	31.7	3.963		
<i>Thalassarche melanophrys</i>	Sex	1	0.172	0.1719	0.218	0.649
	Residuals	12	9.444	0.787		

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Table S7: Linear model estimates for $\delta^{15}\text{N}$ vs. $\log(\text{weight})$ and $\delta^{13}\text{C}$ vs. $\log(\text{weight})$; Estimate, Standard error, F-statistic, t value, p value and adjusted r-square

$\delta^{13}\text{C}$							$\delta^{15}\text{N}$					
Species	Estimate	Std. Error	F	t value	p-val	adj.r ²	Estimate	Std. Error	F	t value	p-val	adj.r ²
<i>Bathyraja albomaculata</i>	-17.172	0.653	2.898	-26.292	<0.001	0.137	10.476	0.627	29.951	16.700	<0.001	0.707
	0.162	0.095		1.702	0.117		0.501	0.091		5.473	<0.001	
<i>Bathyraja brachyurops</i>	-17.536	0.479	2.014	-36.623	<0.001	0.046	10.714	0.673	20.532	15.910	<0.001	0.482
	0.103	0.073		1.419	0.171		0.464	0.103		4.531	<0.001	
<i>Cottoperca gobio</i>	-18.654	0.563	0.926	-33.111	<0.001	-0.004	11.112	0.489	36.367	22.739	<0.001	0.651
	0.098	0.102		0.962	0.349		0.533	0.088		6.031	<0.001	
<i>Dipturus lamillai</i>	-17.743	0.847	0.837	-20.950	<0.001	-0.012	13.486	1.985	0.466	6.793	<0.001	-0.040
	0.096	0.105		0.915	0.377		0.169	0.247		0.683	0.507	
<i>Dissostichus eleginoides</i>	-21.157	1.297	4.264	-16.308	<0.001	0.266	10.261	1.346	6.279	7.626	<0.001	0.370
	0.393	0.190		2.065	0.073		0.494	0.197		2.506	0.037	
<i>Doryteuthis gahi</i>	-19.086	0.216	0.052	-88.178	<0.001	-0.004	11.807	0.256	10.683	46.114	<0.001	0.035
	0.013	0.058		0.227	0.820		0.225	0.069		3.268	0.001	
<i>Genypterus blacodes</i>	-17.417	1.066	0.104	-16.334	<0.001	-0.111	17.694	0.899	2.334	19.684	<0.001	0.129
	0.048	0.147		0.322	0.756		-0.190	0.124		-1.528	0.165	
<i>Illex argentinus</i>	-21.464	0.402	34.179	-53.361	<0.001	0.689	8.648	1.128	14.444	7.667	<0.001	0.473
	0.478	0.082		5.846	<0.001		0.871	0.229		3.801	0.002	
<i>Macruronus magellanicus</i>	-18.203	0.928	0.153	-19.617	<0.001	-0.064	11.836	1.561	1.395	7.581	<0.001	0.027
	0.060	0.154		0.391	0.702		0.307	0.260		1.181	0.259	
<i>Merluccius hubbsi</i>	-21.427	1.172	9.538	-18.281	<0.001	0.437	10.676	1.481	11.381	7.207	<0.001	0.486
	0.538	0.174		3.088	0.011		0.743	0.220		3.374	0.007	

<i>Moroteuthopsis ingens</i>	-19.495	0.866	0.454	-22.500	<0.001	-0.035	9.850	1.497	1.393	6.580	<0.001	0.024
	-0.090	0.134		-0.674	0.511		0.273	0.231		1.180	0.256	
<i>Patagonotothen ramsayi</i>	-20.517	0.280	52.891	-73.333	<0.001	0.604	10.875	0.350	20.333	31.071	<0.001	0.362
	0.455	0.063		7.273	<0.001		0.353	0.078		4.509	<0.001	
<i>Puffinus griseus</i>	-10.860	16.566	0.253	-0.656	0.531	-0.090	44.528	27.566	1.327	1.615	0.145	0.035
	-1.213	2.410		-0.503	0.628		-4.620	4.011		-1.152	0.283	
<i>Salilota australis</i>	-19.707	0.230	67.651	-85.623	<0.001	0.769	11.239	0.605	38.276	18.572	<0.001	0.651
	0.314	0.038		8.225	<0.001		0.622	0.101		6.187	<0.001	
<i>Schroederichthys bivirus</i>	-17.211	0.453	2.760	-38.025	<0.001	0.094	11.950	0.721	9.784	16.575	<0.001	0.341
	0.129	0.078		1.661	0.116		0.387	0.124		3.128	0.006	
<i>Squalus acanthias</i>	-16.036	5.072	0.215	-3.162	0.016	-0.109	15.767	4.986	0.109	3.163	0.016	-0.125
	-0.325	0.701		-0.464	0.657		-0.228	0.689		-0.330	0.751	
<i>Thalassarche melanophrys</i>	2.662	13.291	2.609	0.200	0.845	0.110	15.276	19.470	0.007	0.785	0.448	-0.083
	-2.543	1.574		-1.615	0.132		-0.199	2.306		-0.086	0.933	

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